CSCI 135 Fundamentals of Computer Science I
Module 7 Problem 1

This is an extension of the Module 6 Die Tosses Lab. This is a repeat of the last problem, but you MUST use a dictionary to store the die sides and number of tosses.

Modify your die tosses program to use a Dictionary. Incorporate the following functionality:

1. Prompt the user to enter s, the number of sides on the die. Use error-checking for data type and value within range. Die can have 4-20 sides, and can only have an even number of sides.
2. Prompt the user to enter each side of the die. Each side will be labeled with an integer, a word, or a single character. The side labels will be used as the keys in the dictionary.
3. Create a dictionary where the keys are the side labels, and the values are the number of tosses that landed that side. This dictionary should be initialized with the sides and zero values for the number of tosses.
4. Prompt the user to enter n, the number of tosses. Use error-checking for data type and value within range. The number of tosses should be equal to or larger than the number of sides, and no larger than 1000.
5. Perform the die tosses using randomization. Update the dictionary to store the current number of tosses for each side.
6. Output: each side along with its corresponding number of tosses and percentage of the total tosses.

Constraints: no global variables; die sides and tosses must be stored in a dictionary where sides are the keys and tosses are the values

Hint: there are conversion functions that convert lists to dictionaries, and the like. You must use a dictionary in the way described above. You can also use lists and/or other collections to make coding easier. Be clever and save yourself work!

Please break your program into functions, similar to the last two programs. Your code should be readable and straightforward -- use good comments and variable names. Your functions should be easy to follow and make sense. Write a main function to call the other functions.